system. The need for complex laminations of rotors with their punctured flow barriers makes synchronous reluctance motor rotors difficult to manufacture, somewhat brittle, and therefore not suitable for high-speed operations.

Synchronous reluctance motors are suitable for a wide range of industrial applications that do not require high overloads and high speeds. These are also increasingly being offered for variable speed pumps due to their efficiency.

Their advantages:

- There is no concern of demagnetisation, so these are inherently more reliable than permanent magnet motors
- There need not be any exciting field as torque is zero, thus eliminating electromagnetic spinning losses
- Synchronous reluctance machine rotors can be constructed entirely from high-strength, low-cost materials

Their disadvantages:

- High cost compared to induction motors
- Need speed synchronisation to invertor output frequency by using rotor position sensor and sensorless control
- Compared to induction motor it is slightly heavier and has low power factor. But by increasing the saliency ratio Lds/Lqs, the power factor can be improved

Their applications:

- Metering pumps
- Auxiliary time mechanism
- Wrapping and folding machines
- Proportioning devices on pumps or conveyors
- Synthetic fibre manufacturing equipment
- Processing continuous sheet or film material

Switched reluctance motors

Switched reluctance motors generate torque by magnetically pulling the rotor protrusion to the stator field. However, their stators have relatively few poles. Their rotors are also much simpler with magnetic protrusion due to the toothed profile rather than due to the internal flow barriers in synchronous reluctance motors. The difference in the number of stators and rotor poles causes the Vernier effect and the switched reluctance motors rotor usually rotates in the opposite direction and at a different speed than the stator field.

(a) Stator (b) Rotor

Fig. 6: Parts of a switched reluctance motor

Unlike synchronous reluctance motors, switched reluctance motors typically use DC pulse excitation and require special inverters to operate. A certain current is required to support their magnetic field, leading to lower torque density and efficiency at sweet spots than permanent magnet motors. However, like synchronous reluctance motors, switched reluctance motors are usually one frame size smaller than comparable induction motors.

The great advantage of switched reluctance motors is that field weakening occurs naturally when excitation decreases, without loss of efficiency. As a result, a wide CPSR (greater than 10:1) is easily possible. Efficiency remains good at higher speeds and low loads, and the motors can achieve remarkably constant efficiencies over a wide range of operating conditions.

Switched reluctance motors are also extremely fault tolerant. Without magnets, there is no uncontrolled torque, current under faulty winding conditions, or uncontrolled generation at high speeds. Further, as the switched reluctance motor phases are electrically independent, the motor can, when or if desired, run at reduced power but with higher torque ripple when one or more phases are off. This can be useful if designers want fault tolerance and redundancy.

Thanks to the simple construction of switched reluctance motors, they are durable and undemanding to production. No expensive materials are required and the plain steel rotor is suitable for high speeds and harsh environments. And short-pitch stator coils reduce the risk of short-circuit windings. In addition, the end speed can be short, so the switched reluctance motors are efficiently small and prevent unnecessary stator losses.

The power factor of the switched reluctance motors is lower than the power factor of permanent magnet or induction motors, but the inverter does not need to synthesize a sine wave for efficient motor operation. The switching frequencies of the inverter and the associated switching losses are therefore low.

The main disadvantages of switched reluctance motors are acoustic noise and vibration. These can be controlled by careful mechanical construction, electronic control, and motor design of the application.

Surface rotor magnets (SPM) and internal magnet motors (IPM) are both variants of permanent magnet motors. AC, SPM, and IPM motors show low performance in high-peed range (field attenuation). Stable induction motor capacity is weak unless the motor is powered by an inverter.

Switched reluctance motors are suitable for a wide range of applications and are increasingly used in the handling of heavy materials due to their large moments of rupture and overload. Thanks to their high switched reluctance motor overload capability and wide range of constant power, they are attractive for vehicle traction, not only in off-road equipment but also in cars where they still have to assert themselves. This may be due to concerns about acoustic noise or torque ripple, but it should be borne in mind that an internal combustion motor is successful despite being an excellent source. The use of a switched reluctance motor with a permanent magnet increases the efficiency by several percent compared to an induction motor.

Their advantages:

- Low cost resulting from simple construction
- High reliability
- High fault tolerance
- Heat generated in stator is easy to remove